Notes on Jarda's Remark

April 15, 2026

Abstract

This note constructs a Hamilton–Jacobi–Bellman (HJB) equation for the Mirrlees optimal nonlinear income-tax problem by exploiting a structural parallel with the continuous-time Ramsey tax problem studied in Jiang et al. (2026). In the Ramsey problem, decisions are ordered by time $t \geq 0$ and a single implementability constraint attaches a constant Lagrange multiplier Φ_0 to the planner's problem; the value function is linear in a scalar state variable X_t. In the Mirrlees problem, decisions are ordered by the household's productivity type $z \in [z_0, \bar{z}]$ and a single resource constraint attaches a constant multiplier μ; the appropriate HJB has the value function linear in the state variable $\mathcal{U}(z)$—the utility assigned to type z—under quasilinear preferences. In the general case, the HJB still has a canonical form but the value function is not linear in $\mathcal{U}$. The note draws a correspondence table between the two problems.

1 The Ramsey Problem with No Initial Term Debt

1.1 Environment

Following Jiang et al. (2026), hereafter JSW, consider a continuous-time version of a special case of the Lucas and Stokey (1983) economy in which there is no initial term debt to be financed. A representative household and a benevolent government operate on $t \in [0, \infty)$. The resource constraint is

$$C_t + G_t = N_t, \qquad t \geq 0, \tag{1}$$

where C_t is consumption, N_t is labor supply, and $G_t \geq 0$ is an exogenous government expenditure flow. The household's lifetime utility is

$$\int_0^\infty e^{-\rho t}\, U(C_t, N_t)\, dt, \qquad \rho > 0. \tag{2}$$

Throughout this section we impose the special case:

$$b_{0-,t} = 0 \quad \text{for all } t \geq 0, \tag{3}$$

meaning that the government carries *no* initial term debt.[1]

1.2 The implementability constraint

Using the household's first-order conditions and the government's budget constraint, JSW show that an allocation $\{C_t, N_t\}$ is a competitive equilibrium allocation if and only if it satisfies the resource constraint (1) and the *implementability constraint*. Under (3) this constraint simplifies to

$$\int_0^\infty e^{-\rho t}\big[C_t\, U_{C,t} + N_t\, U_{N,t}\big]\, dt = 0. \tag{4}$$

The Ramsey planner selects the best feasible allocation by maximizing $\int_0^\infty e^{-\rho t} U(C_t, N_t)\, dt$ subject to (1)–(4). The tax rate and competitive equilibrium price system can be deduced from the allocation.

1.3 The cumulative-liability state variable

Because $b_{0-,t} = 0$, the government's cumulative liability Π_t (the net financial position the government would carry if it never restructured its debt) reduces to

$$\Pi_t = \int_0^t \frac{q_{0,s}}{q_{0,t}} \big(G_s - \tau_s N_s\big)\, ds, \qquad \Pi_0 = 0, \tag{5}$$

[1] JSW focus on versions of Lucas and Stokey's Ramsey problems in which a non-zero schedule of initial term debts is central. That part of their analysis is not the focus of this note and would just obscure the connections that we focus on here. JSW altered the timing protocol relative to Lucas and Stokey (1983) in order to overcome the difficulty identified by Debortoli et al. (2021). Jiang et al. (2026) is a continuous-time working paper; a discrete-time companion that incorporates the timing-protocol change but not the dynamic program is forthcoming in the *Journal of Political Economy* as Jiang et al. (2027) We retain JSW's timing protocol here.

where $q_{0,t} = e^{-\rho t} U_{C,t}/U_{C,0}$ is the zero-coupon bond price and τ_t is the flat labor-income tax rate. In differential form,

$$d\Pi_t = r_t \, \Pi_t \, dt + (G_t - \tau_t N_t) \, dt, \tag{6}$$

where $r_t = \rho + \dot{C}_t/C_t$ is the equilibrium short rate.

Remark 1 (Backward-looking character of Π_t)**.** *Equation* (5) *defines* Π_t *as an integral over past primary deficits, making it a purely backward-looking state variable: its value at time* t *is fully determined by the history of* (G_s, τ_s, N_s) *for* $s \leq t$*.*

JSW define the scalar state variable

$$X_t = \Pi_t \, U_{C,t}, \tag{7}$$

which encodes both the government's cumulative liability and the household's marginal utility of consumption. Differentiating and using (6) together with the household's first-order conditions, they obtain

$$\dot{X}_t = \rho \, X_t - C_t \, U_{C,t} - N_t \, U_{N,t}, \tag{8}$$

where the $b_{0-,t}$ terms present in the general JSW formula vanish under (3). Note that the dynamics are *linear* in X_t (the ρX_t term).

Remark 2 (Timing protocol and X_t as a state variable)**.** *Because* $X_t = \Pi_t U_{C,t}$ *depends on the current marginal utility* $U_{C,t}$*, whether it is predetermined requires care. Jiang et al. (2026) adopt the protocol that for all* $t > 0$ *the tax rate at time* t *is set at* $t - dt$*, making* C_t *(and hence* $U_{C,t}$*) predetermined for all* $t > 0$*. This makes* X_t *a genuine state variable for the continuation problem at* $t > 0$*, and is the continuous time instantiation of JSW's key modification of Lucas and Stokey's original timing protocol.*

Remark 3. *(Time* 0 *is different) The Ramsey planner chooses* C_0 *at time* 0*. This makes the Ramsey planner's decision problem different from the decision problems of the "continuation Ramsey planners" to be introduced soon.*

Initial condition. Because $\Pi_0 = 0$, the state starts at $X_0 = 0$.

1.4 The HJB equation

Define a value function

$$V(X_t, t) = \max_{\{C_s;\, s \geq t\}} \int_t^\infty e^{-\rho(s-t)} U(C_s, C_s + G_s)\, ds, \tag{9}$$

where maximization is subject to the law of motion (8). By the principle of dynamic programming, V satisfies the HJB equation

$$\rho V = V_t + \max_{C_t} \Big\{ U(C_t, C_t + G_t) + V_X \left(\rho X_t - C_t U_{C,t} - N_t U_{N,t} \right) \Big\}. \tag{10}$$

The optimization problem on the right side of (10) is that of a *continuation Ramsey planner* at time t who faces state variable X_t and chooses C_{t+dt}.

1.5 The Ramsey planner and the continuation Ramsey planners

Following Lucas and Stokey (1983) and the QuantEcon lectures on optimal taxation,[2] it is instructive to distinguish sharply between two roles in recursive implementations of a Ramsey plan, each with its own state and choice variables.

The Ramsey planner at $t = 0$. The Ramsey planner operates once, at $t = 0$.

- *State variable.* There is no initial term debt for the Ramsey planner to service; consequently, it faces state variable $X_0 = 0$ (see (3)).
- *Choice variables.* (i) The time-0 allocation (C_0, N_0); and (ii) the entire trajectory $\{X_s\}_{s>0}$ of promised values to be bequeathed to continuation planners at every future instant. By the structure of the HJB these choices collapse to a single free scalar: the Lagrange multiplier $\Phi_0 \geq 0$ on (4). Once Φ_0 is chosen, the policy rule $C^* = C(\Phi_0; G_t)$ from (13) pins down consumption at every $t \geq 0$.
- *Value function.* $V(0, 0; \Phi_0^*) = H(0; \Phi_0^*)$, since $X_0 = \Pi_0 U_{C,0} = 0$.

A continuation Ramsey planner at $t > 0$. A continuation Ramsey planner arrives at time $t > 0$ with an inherited state.

[2] See `https://python-advanced.quantecon.org/opt_tax_recur.html` and `https://python-advanced.quantecon.org/amss.html`

- *State variables.* (X_t, t), where $X_t = \Pi_t U_{C,t}$ is the marginal-utility-scaled cumulative liability. The continuation planner is *obligated* to take X_t as given.
- *Choice variable.* Current consumption C_t. Through the law of motion (8), choosing C_{t+dt} implies the promised value X_{t+dt} passed to the next continuation planner.
- *Value function.* $V(X_t, t; \Phi_0) = -\Phi_0 X_t + H(t; \Phi_0)$. The shadow price $V_X = -\Phi_0$ is the *same constant* for every continuation planner along the original Ramsey path.

The time-inconsistency signature. The two planners differ not only in their state variables but in the constraints that determine their first-order conditions. The time-0 Ramsey planner faces initial state $X_0 = 0$ and chooses C_0 and Φ_0; a continuation Ramsey planner at $t > 0$ inherits the state $X_t = \Pi_t U_{C,t}$ and chooses C_{t+dt}.

1.6 Linear value-function guess and verification

Guess. Try the additively separable form

$$V(X_t, t;\, \Phi_0) = -\Phi_0\, X_t + H(t;\, \Phi_0), \tag{11}$$

where $\Phi_0 \geq 0$ is a scalar and H is a function of time and Φ_0. Then $V_X = -\Phi_0$ (constant in both X_t and t).

Verification. Substituting (11) into (10), the $-\Phi_0\, \rho X_t$ terms on both sides cancel, and the HJB reduces to the following ODE for H:

$$\rho\, H(t;\, \Phi_0) = H'(t;\, \Phi_0) + \max_{C_t} \Big\{ U(C_t, C_t + G_t) + \Phi_0 \big(C_t U_{C,t} + N_t U_{N,t} \big) \Big\}. \tag{12}$$

The maximization over C_t on the right side is independent of the state X_t. Optimal consumption satisfies

$$C(\Phi_0;\, G_t) = \arg\max_{C} \Big\{ U(C, C + G_t) + \Phi_0 \big(C\, U_C + (C + G_t) U_N \big) \Big\}. \tag{13}$$

1.7 Determining the multiplier Φ_0

Substituting the optimal plan $C^* = C(\Phi_0; G_t)$ and $N^* = C^* + G_t$ into the implementability constraint (4) and imposing $X_0 = 0$, the Ramsey multiplier Φ_0^* is the

non-negative root of

$$I_0(\Phi_0) \equiv \int_0^\infty e^{-\rho t}\big[C(\Phi_0;\, G_t)\, U_C + N(\Phi_0;\, G_t)\, U_N\big]\, dt = 0, \tag{14}$$

where $N(\Phi_0; G_t) = C(\Phi_0; G_t) + G_t$.

Remark 4 (Interpretation of Φ_0)**.** *The multiplier Φ_0 is the shadow price of the single implementability constraint and is* constant across all dates*. This constancy reflects the exponential discount structure in the Ramsey problem: the ρX_t term in the law of motion* (8) *exactly cancels the discounting on the left side of the HJB, keeping the shadow price $-V_X = \Phi_0$ time-invariant.*

1.8 Cyclical government expenditures

To illustrate the power of the linear guess, we now specialise to $U(C, N) = \ln C - N$ (the baseline utility used in JSW's quantitative examples) and allow government spending to oscillate around a mean:

$$G_t = G_0 + \theta_G \cos(\omega t), \qquad G_0,\, \theta_G > 0,\ \omega > 0, \tag{15}$$

so that $G_t > 0$ for all t provided $\theta_G < G_0$.

1.8.1 Constant optimal tax from the HJB

Under $U = \ln C - N$ we have $U_C = 1/C$ and $U_N = -1$. With $b_{0-,t} = 0$, the maximization problem (13) in the HJB becomes

$$C(\Phi_0;\, G_t) = \arg\max_{C>0}\Big\{\ln C - (C+G_t) + \Phi_0\big[C\cdot\tfrac{1}{C} + (C+G_t)(-1)\big]\Big\} = \arg\max_{C>0}\Big\{\ln C - (1+\Phi_0)(C+G_t)\Big\}. \tag{16}$$

The first-order condition $1/C = 1 + \Phi_0$ yields

$$C^* = \frac{1}{1+\Phi_0^*}, \qquad \tau^* = 1 - C^* = \frac{\Phi_0^*}{1+\Phi_0^*}. \tag{17}$$

Crucially, C^* and τ^* are *independent of* G_t: the cyclical component of fiscal needs does not appear in the optimality condition for consumption. The Ramsey planner therefore imposes a *constant* tax rate at every date, even though government expenditures oscillate. This is the HJB incarnation of the tax-smoothing result of Barro (1979).

Remark 5 (Why G_t drops out)**.** *The HJB maximand contains G_t twice: once in the felicity term $-(C + G_t)$ and once in the incentive term $-(1 + \Phi_0)G_t$. Both enter additively as constants in the maximization over C. The optimum C^* is therefore pinned by the ratio of marginal utility to the Lagrange multiplier, with no role for G_t.*

1.8.2 Determining Φ_0^* from the implementability constraint

The implementability constraint (14) with $U_C = 1/C^*$ and $U_N = -1$ reduces to

$$\int_0^\infty e^{-\rho t}\big[1 - (C^* + G_t)\big]\, dt = 0, \tag{18}$$

which gives

$$\frac{1 - C^*}{\rho} = \int_0^\infty e^{-\rho t} G_t\, dt = \frac{G_0}{\rho} + \theta_G \frac{\rho}{\rho^2 + \omega^2}. \tag{19}$$

Hence

$$1 - C^* = G_0 + \theta_G \frac{\rho^2}{\rho^2 + \omega^2}. \tag{20}$$

Three observations follow. First, the equilibrium tax rate depends only on the discounted present value of government expenditures, not on their timing. Second, the cyclical component is attenuated by $\rho^2/(\rho^2 + \omega^2) \in (0, 1)$: high-frequency cycles ($\omega \gg \rho$) contribute negligibly to the tax rate because they nearly cancel in present value. Third, as $\theta_G \to 0$ the formula collapses to $1 - C^* = G_0$.

1.8.3 The cyclical state variable X_t^*

Even though C^* and τ^* are constant, the state variable $X_t^* = \Pi_t^* U_{C,t}^* = \Pi_t^*/C^*$ inherits the cyclicality of G_t. Substituting $C^* = \text{const}$, $N_t^* = C^* + G_t$ into the law of motion (8):

$$\dot{X}_t = \rho X_t - 1 + C^* + G_t = \rho X_t - \frac{\theta_G \rho^2}{\rho^2 + \omega^2} + \theta_G \cos(\omega t), \tag{21}$$

where we used $1 - C^* = G_0 + \theta_G \rho^2/(\rho^2 + \omega^2)$ to absorb G_0. Solving with $X_0 = 0$:

$$X_t^* = \frac{\theta_G}{\rho^2 + \omega^2}\big[\rho\big(1 - \cos(\omega t)\big) + \omega \sin(\omega t)\big]. \tag{22}$$

This solution is bounded, oscillates with period $2\pi/\omega$, and starts near zero. For $\omega \gg \rho$ the amplitude is approximately θ_G/ω, so high-frequency spending shocks produce only small oscillations in the state variable.

Remark 6 (X_t^* as a fiscal buffer)**.** *Variations in X_t^* map one-for-one into variations in Π_t^* (since C^* is constant). When G_t is high the government runs a deficit, pushing Π_t^* up; when G_t falls back, the surplus rises and Π_t^* declines. The tax rate is constant throughout because it is the* state variable*, not the tax rate, that absorbs fiscal fluctuations.*

1.8.4 Structure of $H(t;\, \Phi_0^*)$

The time-varying component $H(t;\, \Phi_0^*)$ satisfies the ODE (12) with $C_t = C^*$ fixed:

$$\rho H(t;\, \Phi_0^*) = H'(t;\, \Phi_0^*) + \ln C^* - (C^* + G_t) + \Phi_0^* \big[1 - (C^* + G_t)\big]. \tag{23}$$

Because G_t oscillates, H has the form

$$H(t;\, \Phi_0^*) = H_\infty + A_H \cos(\omega t) + B_H \sin(\omega t), \tag{24}$$

where $H_\infty = [\ln C^* - (1 + \Phi_0^*)C^*]/\rho$ and (A_H, B_H) are determined by matching ω-frequency terms. The value function $V(X_t^*, t) = -\Phi_0^* X_t^* + H(t; \Phi_0^*)$ oscillates both through the state X_t^* and directly through $H(t)$, together capturing the welfare cost of the cyclical fiscal program. The optimal control rule is time-invariant throughout: the HJB linear guess cleanly separates the time-varying fiscal environment (absorbed by H) from the constant policy rule (pinned by Φ_0^*).

2 The Mirrlees Problem: A Sequential Structure in Increasing Unobserved Productivity

2.1 Environment

Following the lecture notes of Bhandari (2026) on optimal nonlinear taxation, there is a continuum of households indexed by productivity $z \in \mathcal{Z} = [z_0, \bar{z}]$ with density $f(z) > 0$. A type-z household produces output

$$y(z) = z\, n(z), \tag{25}$$

where $n(z)$ is labor and z is the marginal product. Utility over consumption c and labor n is $U(c, n)$. The government observes only income y, not the decomposition

(z, n).

2.2 The mechanism design formulation

By the taxation principle, the government can equivalently optimize over incentive-compatible (IC) allocations $\{c(z), y(z)\}$. Defining household utility

$$\mathcal{U}(z) = U\left(c(z),\ \frac{y(z)}{z}\right), \tag{26}$$

the local IC condition (envelope theorem applied to the announcement problem) is

$$\frac{d\mathcal{U}}{dz} = -U_n\left(c(z),\ \frac{y(z)}{z}\right)\frac{y(z)}{z^2} \equiv \phi(z), \tag{27}$$

where $\phi(z) > 0$ because $U_n < 0$. Condition (27) is the *envelope ODE*: it plays the role of the law of motion for the state variable when the problem is viewed sequentially in z.

Consumption is the implicit function of state and control:

$$c(z) = C\big(\mathcal{U}(z),\ y(z)\big), \qquad \text{where } \mathcal{U} = U(C(\mathcal{U}, y),\ y/z). \tag{28}$$

Its partial derivatives are $C_{\mathcal{U}} = 1/U_c$ and $C_y = -U_n/(zU_c)$.

2.3 The planner's problem

The government maximizes social welfare subject to the IC law of motion (27) and a resource constraint:

$$\max_{y(\cdot),\mathcal{U}(z_0)} \int_{z_0}^{\bar{z}} \psi(z)\,\mathcal{U}(z)\,f(z)\,dz \tag{29}$$

$$\text{s.t.} \qquad \frac{d\mathcal{U}}{dz} = -U_n\big(C(\mathcal{U}, y),\ y/z\big)\frac{y}{z^2}, \tag{30}$$

$$\int_{z_0}^{\bar{z}} \big[y(z) - C(\mathcal{U}(z), y(z))\big]\,f(z)\,dz \geq B. \tag{31}$$

The control is the output schedule $y(\cdot)$; the state is $\mathcal{U}(z)$. The initial utility level $\mathcal{U}(z_0)$ is a free endpoint to be optimized.

The single *global* constraint (31) plays the same structural role as the implementabil-

ity constraint (4) in the Ramsey problem. Attaching a non-negative Lagrange multiplier μ to (31), the augmented objective is

$$\max_{y(\cdot),\mathcal{U}(z_0)} \int_{z_0}^{\bar{z}} \big[\psi(z)\,\mathcal{U}(z) + \mu\big(y(z) - C(\mathcal{U}(z), y(z))\big)\big]\, f(z)\, dz - \mu B. \tag{32}$$

3 The Mirrlees HJB Equation

3.1 The sequential value function

Viewing productivity z as the "index" along which the planner proceeds (just as t indexes the Ramsey planner's decisions), define the *continuation value* for the planner who has already assigned allocations to types below z and now faces types $\zeta > z$ with inherited state $\mathcal{U}(z) = u$:

$$V(u, z;\, \mu) = \max_{y(\cdot)} \int_{z}^{\bar{z}} \big[\psi(\zeta)\,\mathcal{U}(\zeta) + \mu\big(y(\zeta) - C(\mathcal{U}(\zeta), y(\zeta))\big)\big]\, f(\zeta)\, d\zeta, \tag{33}$$

subject to (30) and $\mathcal{U}(z) = u$.

Sequential decision-making protocol and promised values. The state variable $u = \mathcal{U}(z)$ is the utility that has already been *promised* to type z by the assignments made to all lower types $\zeta < z$. A continuation planner starting at z is obliged to deliver exactly u to type z. Types already processed ($\zeta < z$) have received their allocations and cannot be changed. This is the Mirrlees analogue of the commitment structure in the Ramsey problem: just as $X_t = \Pi_t U_{C,t}$ records the cumulative fiscal obligation inherited by each continuation Ramsey planner, the promised utility u records the incentive obligation inherited by each continuation Mirrlees planner.

Remark 7. *(Productivity z_0 is different) The Mirrlees planner at $z = z_0$ chooses $y(z_0)$ and the initial promised utility $u(z_0)$. This makes the z_0-planner's decision problem different from those of the "continuation Mirrlees planners" to be introduced soon.*

3.2 The Mirrlees planner and the continuation Mirrlees planners

Paralleling the Ramsey treatment in Section 1.5, we distinguish the Mirrlees planner at the bottom of the type distribution from all continuation Mirrlees planners.

The Mirrlees planner at $z = z_0$. The planner at z_0 faces no inherited commitment.

- *State variable.* The initial promised utility $u(z_0)$, which is a *free endpoint* in problem (29); no prior type has imposed an obligation. The participation constraint pins a floor on $u(z_0)$, playing the role that the initial government debt plays in the general Ramsey problem (set to zero here under (3)).

- *Choice variables.* (i) The output $y(z_0)$ for the lowest type; and (ii) the entire schedule $\{\mathcal{U}(\zeta)\}_{\zeta > z_0}$ of promised utilities to bequeath to continuation planners. By the structure of the HJB these choices collapse to the single scalar $\mu \geq 0$ (the Lagrange multiplier on the resource constraint (31)): once μ is set, the FOC (40) pins down output at every type.

- *Value function.* $V(u(z_0), z_0; \mu) = \lambda(z_0)\, u(z_0) + K(z_0; \mu)$. The boundary condition $\lambda(z_0) = 0$ means the z_0-planner's value reduces to $K(z_0; \mu)$, the aggregate resource surplus from the entire mechanism.

The continuation Mirrlees planner at $z > z_0$. A continuation Mirrlees planner at z arrives with an inherited state.

- *State variables.* (u, z), where $u = \mathcal{U}(z)$ is the promised utility inherited from all types below z. The continuation planner is *obligated* to honor u and cannot renegotiate it.

- *Choice variable.* The output $y(z)$ for type z. Through the IC law of motion (30), choosing $y(z)$ implies the promised utility $\mathcal{U}(z + dz)$ passed to the next continuation planner.

- *Value function.* $V(u, z; \mu) = \lambda(z)\, u + K(z; \mu)$. The shadow price $V_u = \lambda(z)$ records the social cost of tightening incentive constraints at type z.

Constant versus varying shadow prices. A key difference from the Ramsey problem is that the Mirrlees shadow price $V_u = \lambda(z)$ *varies* along the type distribution, whereas the Ramsey shadow price $-V_X = \Phi_0$ is *constant* across all continuation planners. The Mirrlees costate $\lambda(z)$ satisfies the ODE (38) and changes because the redistribution pressure shifts as the mechanism moves through the type distribution: $\lambda(z)$ is typically hump-shaped, large (in absolute value) where the planner most values relaxing the incentive constraint, and zero at both boundaries. The constancy of Φ_0 in

Ramsey reflects the homogeneity of fiscal policy across time under commitment; the variation of $\lambda(z)$ in Mirrlees reflects the heterogeneity of incentive costs across types.

3.3 The HJB equation

By the principle of dynamic programming, V satisfies

$$-\frac{\partial V}{\partial z} = \max_y \Big\{ \big[\psi(z)\, u + \mu\big(y - C(u,y)\big)\big] f(z) + \frac{\partial V}{\partial u} \cdot \Big(-U_n\big(C(u,y),\, y/z\big) \frac{y}{z^2} \Big) \Big\}. \quad (34)$$

This is the *Mirrlees HJB equation.* Writing $\phi(u,y,z) \equiv -U_n(C(u,y), y/z)\, y/z^2 > 0$, (34) can be compactly stated as

$$-V_z = \max_y \Big\{ [\psi(z)\, u + \mu(y - C(u,y))] f(z) + V_u \cdot \phi(u,y,z) \Big\}. \quad (35)$$

Boundary condition. At the top of the type distribution there is no continuation value, so

$$V(u, \bar{z};\, \mu) = 0 \quad \text{for all } u. \quad (36)$$

3.4 The linear value-function guess

Guess. Motivated by the Ramsey analogy, try

$$V(u, z;\, \mu) = \lambda(z;\, \mu) \cdot u + K(z;\, \mu), \quad (37)$$

where λ is a function of z and K is a separate function of z and μ. Then $V_u = \lambda(z)$.

When does the guess work? The linear guess is valid whenever the maximand in (35) is *affine in* u. This requires:

(i) The flow payoff $[\psi(z)u + \mu(y - C(u,y))]f(z)$ to be affine in u, which holds when $C(u,y)$ is affine in u.

(ii) The state dynamics $\phi(u,y,z)$ to be independent of u (or affine in u), which holds when $U_n(C(u,y), y/z)$ is independent of u.

Both conditions are satisfied under *quasilinear preferences*, $U(c,n) = c - v(n)$, since then $C(u,y) = u + v(y/z)$, which is linear in u with slope 1, and $U_n = -v'(y/z)$ is independent of c and hence of u.

Verification under quasilinear preferences. With $U = c - v(n)$:

$$C(u, y) = u + v(y/z),$$
$$\phi(u, y, z) = v'(y/z)\, y/z^2.$$

Substituting the linear guess (37) into (35):

$$-\lambda'(z)\, u - K'(z)$$
$$= \max_y \Big\{ \big[\psi(z) - \mu\big] u\, f(z) + \mu\big[y - v(y/z)\big] f(z) + \lambda(z)\, v'(y/z)\, y/z^2 \Big\}.$$

Separating terms in u from terms independent of u:

$$\lambda'(z) = \big[\mu - \psi(z)\big] f(z), \tag{38}$$

$$K'(z) = -\max_y \Big\{ \mu\big[y - v(y/z)\big] f(z) + \lambda(z)\, v'(y/z)\, y/z^2 \Big\}. \tag{39}$$

Equation (38) is exactly the costate equation derived from the Hamiltonian approach in the lecture notes.

3.5 Optimality condition for the output schedule

The first-order condition for the maximization problem in (39) (in the quasilinear case) is

$$0 = \mu\big[1 - v'(y/z)/z\big] f(z) + \frac{\lambda(z)}{z^2}\big[v'(y/z) + n\, v''(y/z)\big], \tag{40}$$

where $n = y/z$. Rearranging using $v'(n) = z(1 - T'(y))$:

$$\frac{T'(y(z))}{1 - T'(y(z))} = \Big[1 + \frac{n\, v''(n)}{v'(n)}\Big] \cdot \frac{\lambda(z)}{z\, f(z)\, \mu}. \tag{41}$$

This is the Diamond (1998) ABC formula.

3.6 Boundary conditions and determination of μ

At the top. The boundary condition (36) with the guess (37) gives

$$\lambda(\bar{z}) = 0, \qquad K(\bar{z}) = 0. \tag{42}$$

At the bottom. The initial utility $\mathcal{U}(z_0)$ is a free endpoint (the planner can choose the utility of the lowest type). The transversality condition at the free endpoint is $V_u(u, z_0) = 0$, which by the guess (37) gives

$$\lambda(z_0) = 0. \tag{43}$$

Together, (42)–(43) are the two-point boundary conditions on the costate ODE (38). They are exactly the transversality conditions $\lambda(z_0) = \lambda(\bar{z}) = 0$ derived in the lecture notes. In the quasilinear case, integrating (38) from z to $\bar{z}$ using $\lambda(\bar{z}) = 0$ gives

$$\lambda(z) = \int_z^{\bar{z}} \big[\psi(\zeta) - \mu\big] f(\zeta)\, d\zeta, \tag{44}$$

and the lower boundary condition $\lambda(z_0) = 0$ pins down $\mu = 1$ (the marginal utility of public funds equals one under quasilinear preferences).

Determining μ in general. For general preferences, the multiplier μ is determined by the resource constraint (31):

$$\int_{z_0}^{\bar{z}} \big[y^*(z) - C(\mathcal{U}^*(z), y^*(z))\big] f(z)\, dz = B, \tag{45}$$

which plays the role of equation (14) in the Ramsey problem.

4 Structural Correspondence between the Two Problems

Table 1 displays the element-by-element structural parallel between the Ramsey problem (no initial term debt) and the Mirrlees problem.

A key structural difference deserves emphasis:

Remark 8 (Constant vs. varying shadow price)**.** *In the Ramsey problem, the shadow price $V_X = -\Phi_0$ is the same constant at every date t. This constancy arises because the state dynamics* (8) *contain a ρX_t term that cancels the discounting ρV on the left of the HJB* (10)*, leaving the costate equation trivially satisfied with $\Phi_0 = const$.*

In the Mirrlees problem, the shadow price $\lambda(z) = V_{\mathcal{U}}$ evolves with z according to the costate ODE (38)*. The reason is that the envelope ODE* (27) *contains no term*

Table 1: Structural Correspondence: Ramsey vs. Mirrlees

Ramsey problem (no initial debt)	**Mirrlees problem**
Sequential index: time $t \in [0, \infty)$	Sequential index: productivity $z \in [z_0, \bar{z}]$
State variable: $X_t = \Pi_t U_{C,t}$	State variable: $\mathcal{U}(z)$ (household utility)
Control: consumption C_t	Control: output schedule $y(z)$
Law of motion: $\dot{X}_t = \rho X_t - C_t U_{C,t} - N_t U_{N,t}$	Envelope ODE: $\mathcal{U}'(z) = -U_n(c, y/z)\, y/z^2$
Initial condition: $X_0 = 0$	Free endpoint: $\mathcal{U}(z_0)$ optimized; $\lambda(z_0) = 0$
Terminal condition: transversality as $t \to \infty$	Terminal condition: $\lambda(\bar{z}) = 0$
Objective: $\int_0^\infty e^{-\rho t} U(C_t, N_t)\, dt$	Objective: $\int_{z_0}^{\bar{z}} \psi(z)\, \mathcal{U}(z)\, f(z)\, dz$
Single global constraint: implementability $(= 0)$	Single global constraint: resource $(= B)$
Global-constraint multiplier Φ_0	Global-constraint multiplier μ
Value function: $V = -\Phi_0 X_t + H(t;\, \Phi_0)$	Value function: $V = \lambda(z)\mathcal{U} + K(z;\, \mu)$ (linear in $\mathcal{U}$ under quasilinear preferences)
HJB shadow price: $V_X = -\Phi_0$ *(constant)*	HJB shadow price: $V_{\mathcal{U}} = \lambda(z)$ *(varies with z)*
Costate equation: $V_X = -\Phi_0$ constant; $\dot{V}_X = 0$	Costate ODE: $\lambda'(z) = [\mu - \psi(z)] f(z)$
Optimal control: $C(\Phi_0;\, G_t) = \arg\max_C [\ldots]$	Optimal control: $y(z) = \arg\max_y [\ldots]$ (tax formula)
Determines Φ_0: $I_0(\Phi_0) = 0$	Determines μ: resource constraint $= B$

proportional to $\mathcal{U}$ (analogous to no "z-discounting"), and the welfare weight $\psi(z)$ varies with z. The costate $\lambda(z)$ accumulates the gap between the social welfare weight ψ and the social cost of public funds μ, measuring the "redistribution pressure" emanating from types above z.

Remark 9 (Quasilinear preferences as the exact Ramsey analogue)**.** *In the Ramsey problem, the linear value-function guess $V = -\Phi_0 X_t + H(t)$ works for* all *admissible preferences. This is because the dynamics* (8) *are linear in X_t for any preferences, and the optimization over C_t in the HJB is independent of X_t.*

In the Mirrlees problem, the analogous linear guess $V = \lambda(z)\mathcal{U} + K(z)$ works only under quasilinear preferences *$U = c - v(n)$, because only then is the implicit function $C(\mathcal{U}, y) = \mathcal{U} + v(y/z)$ linear in $\mathcal{U}$, making the HJB affine in the state $\mathcal{U}$. For general preferences, $C(\mathcal{U}, y)$ is nonlinear in $\mathcal{U}$, and the value function $V(u, z)$ is nonlinear in u. The HJB* (35) *remains the correct dynamic-programming equation in both cases; it is only the validity of the* linear *guess that is special to quasilinear preferences.*

Remark 10 (Initial planners versus continuation planners)**.** *The table records a further structural parallel: the special role of the* initial *planner in each problem.*

Ramsey. *The entry "Initial condition: $X_0 = 0$" in the table is more than a technicality. The entire Ramsey program is determined at time $t = 0$ by the single number Φ_0^*: once it is chosen, the continuation policy $C^*(t)$, $N^*(t)$, and the state trajectory X_t^* are pinned down for all $t > 0$. The time-0 Ramsey planner is therefore in a unique position — it is the only planner that can set Φ_0 freely (subject to the implementability constraint* (14)*). Every continuation Ramsey planner inherits $X_t^* \neq 0$ and must treat Φ_0 as a given parameter. A continuation Ramsey plan that restarts from some (X_t, t) with $X_t > 0$ is* not *a Ramsey plan starting from t: it would choose a different multiplier $\widetilde{\Phi}(X_t) \neq \Phi_0$, which is precisely the time-inconsistency of the Ramsey plan.*

Mirrlees. *The entry "Free endpoint: $\mathcal{U}(z_0)$ optimized; $\lambda(z_0) = 0$" has the same structure. The entire mechanism is determined at $z = z_0$ by the single number $u(z_0)$: this initial promised utility, together with the Envelope ODE* (30)*, pins down the complete allocation $\{\mathcal{U}(z)\}_{z \geq z_0}$. The planner at z_0 is the only planner free to choose $u(z_0)$ (subject to the participation constraint). A continuation Mirrlees planner that restarts at an intermediate type $z' > z_0$ inherits $u(z') \neq u(z_0)$ and faces a different problem. Commitment by the z_0-planner — to honor the utilities generated by the mechanism for all $\zeta \in [z_0, z']$ — prevents such reoptimization, exactly as commitment by the time-0*

Ramsey planner prevents resetting Φ_0 at $t > 0$.

Summary. *In both problems the initial planner simultaneously (i) faces no inherited state to which it is obligated, and (ii) sets the single free parameter (Φ_0 or $u(z_0)$) that governs the entire continuation. All subsequent planners are bound by a state they cannot revise. The table's "determines Φ_0: $I_0(\Phi_0) = 0$" and "determines μ: resource constraint = B" rows reflect how each problem closes: just as B pins μ through the resource constraint for the Mirrlees mechanism, the implementability constraint closes the Ramsey problem and pins Φ_0^*.*

5 The Mirrlees HJB in the General Case

For general preferences, the HJB (35) still governs the value function, but V is a non-linear function of u. The optimality conditions derived from the Hamiltonian approach of Bhandari (2026)—the $H_y = 0$ condition and the costate equation—are exactly the first-order conditions of the HJB: they arise from differentiating the maximand in (35) with respect to y and equating the z-derivative of $V_{\mathcal{U}}$ to minus the $\mathcal{U}$-gradient of the maximand.

Specifically, denote $\lambda(z) \equiv V_u(u^*(z), z)$ along the optimal path, where $u^*(z) = \mathcal{U}^*(z)$ is the optimal utility assignment. Then:

(i) **Optimality in** y (from $\partial/\partial y$ of the maximand):

$$\mu\,[1 - C_y]f(z) - \lambda(z)\Big[U_{nc}\,C_y + U_{nn}/z\Big]\frac{y}{z^2} - \lambda(z)\frac{U_n}{z^2} = 0. \tag{46}$$

(ii) **Costate equation** (from $-\partial/\partial u$ of the maximand):

$$\frac{d\lambda}{dz} = \mu\,C_{\mathcal{U}}\,f(z) - \psi(z)\,f(z) + \lambda(z)\,U_{nc}\,\frac{y}{z^2}\,C_{\mathcal{U}}. \tag{47}$$

These are identical to equations (5.2) and (5.3) of the lecture notes. The HJB perspective makes clear that $\lambda(z)$ is not an auxiliary variable of the Hamiltonian approach but is the *slope of the value function* $\lambda(z) = V_u(\mathcal{U}^*(z), z)$ evaluated along the optimal path.

6 Summary and Interpretation

Both the Ramsey and Mirrlees planning problems share the following structural architecture:

(1) **A sequential ordering principle.** In Ramsey, decisions are made along the time axis t. In Mirrlees, the social planner processes households from $z = z_0$ upward through higher productivities. In both cases, the planner integrates a *flow payoff* and a *law of motion* for a one-dimensional state variable.

(2) **A single global constraint that attaches a constant multiplier.** In Ramsey, the implementability constraint (4) attaches the constant Φ_0. In Mirrlees, the resource constraint (31) attaches the constant μ. In both cases, this multiplier folds the global constraint into the flow payoff and turns the problem into a standard calculus-of-variations problem (discounted in the Ramsey problem, undiscounted in the Mirrlees problem).

(3) **A Hamilton–Jacobi–Bellman equation.** Once the global constraint is Lagrangianized, the problem admits a value function that satisfies an HJB. The shadow price of the state variable ($V_X = -\Phi_0$ in Ramsey; $V_{\mathcal{U}} = \lambda(z)$ in Mirrlees) is the costate variable of the underlying optimal control problem.

(4) **A determining equation for the global multiplier.** The multiplier is found by imposing the global constraint: $I_0(\Phi_0) = 0$ in Ramsey, and the resource constraint (45) in Mirrlees.

The deepest difference is that in Ramsey the costate is constant (because exponential discounting with rate ρ exactly offsets the ρX_t drift in the state), while in Mirrlees the costate $\lambda(z)$ varies, accumulating the difference between the planner's welfare weights and the social cost of funds. Under quasilinear preferences the linear value-function guess works in both problems.

A The Hamiltonian and HJB for Mirrlees: Equivalence

The Hamiltonian approach of Bhandari (2026) defines

$$\mathcal{H}(\mathcal{U}, y, \lambda;\, z) = \big[\psi(z)\,\mathcal{U} + \mu(y - C(\mathcal{U}, y))\big] f(z) - \lambda\, U_n\big(C(\mathcal{U}, y), y/z\big)\,\frac{y}{z^2}. \tag{48}$$

The HJB (35) is the Hamilton–Jacobi equation

$$-V_z(u, z) = \mathcal{H}\big(u,\, y^*(u, z),\, V_u(u, z);\, z\big), \tag{49}$$

where $y^*(u, z) = \arg\max_y \mathcal{H}(u, y, V_u(u, z); z)$. Along the optimal path, the costate $\lambda(z) = V_u(\mathcal{U}^*(z), z)$ satisfies $d\lambda/dz = -\mathcal{H}_{\mathcal{U}}$ (equation (47) above), confirming that the Hamiltonian and HJB approaches are equivalent.

B The Mirrlees HJB with Pareto-Distributed Productivity

B.1 Setup and motivation

The canonical Mirrlees analysis with bounded support $\mathcal{Z} = [z_0, \bar{z}]$ yields a *zero top marginal tax rate*: $T'(y(\bar{z})) = 0$. This result follows directly from the boundary condition $\lambda(\bar{z}) = 0$ together with the tax formula (41): at the top, $(1 - F(\bar{z})) = 0$ so the costate is zero and the distortion vanishes. A large empirical literature documents that the upper tail of the earnings distribution is well-approximated by a Pareto distribution, not a distribution with bounded support. Diamond and Saez (2011) argue that the zero-top-rate result is an artifact of bounded support and derive a formula for the optimal *limiting* marginal tax rate under an unbounded Pareto tail.

This section extends the HJB framework to unbounded support $\mathcal{Z} = [z_0, \infty)$ and derives the Diamond–Saez top-rate formula directly from the asymptotic behavior of the Mirrlees HJB.

B.2 The Pareto distribution

Let productivity z follow a Pareto distribution with shape $\alpha > 1$ and lower bound z_0:

$$f(z) = \alpha\, z_0^{\alpha}\, z^{-\alpha-1}, \qquad 1 - F(z) = \left(\frac{z_0}{z}\right)^{\alpha}, \qquad z \geq z_0. \tag{50}$$

The inverse hazard rate is

$$\frac{1 - F(z)}{z\, f(z)} = \frac{1}{\alpha} \quad \text{for all } z \geq z_0, \tag{51}$$

a constant. Under the optimal tax schedule, earnings $y(z)$ are also Pareto in the tail. With quasilinear preferences $U = c - v(n)$ and constant elasticity $v(n) = n^{1+1/e}/(1 + 1/e)$, the earnings Pareto parameter is

$$\alpha_y = \frac{\alpha\, e}{1 + e}, \tag{52}$$

where $e = 1/\gamma$ is the compensated labor-supply (or taxable-income) elasticity.

B.3 Transversality replaces the upper boundary condition

With $\bar{z} = \infty$, there is no upper type to impose $\lambda(\infty) = 0$ as a point condition. Instead, optimality requires the transversality condition

$$\lim_{z \to \infty} V(u,\, z;\, \mu) = 0 \quad \text{for all } u, \tag{53}$$

meaning no continuation value remains once all types have been assigned allocations. Under the linear guess $V = \lambda(z)\, u + K(z; \mu)$, this decomposes into

$$\lambda(\infty) = 0, \qquad K(\infty) = 0. \tag{54}$$

These are exactly the conditions that ensure the costate integral (44) converges. For the Pareto distribution, the integrand $[\psi(\zeta) - \mu] f(\zeta)$ decays as $\zeta^{-\alpha-1}$, fast enough for $\alpha > 1$, so the integral converges and $\lambda(\infty) = 0$ is automatically satisfied as a transversality condition rather than imposed as a point constraint.

B.4 Asymptotic costate and the nonzero top rate

Specialize to quasilinear preferences ($\mu = 1$) and assume the planner places zero welfare weight on the top: $\psi(z) \to 0$ as $z \to \infty$ (a purely redistributive planner). For z large enough that $\psi(z) = 0$:

$$\lambda(z) = \int_z^\infty \big[\psi(\zeta) - 1\big] f(\zeta)\, d\zeta \;\approx\; -\int_z^\infty f(\zeta)\, d\zeta = -(1 - F(z)) = -\Big(\frac{z_0}{z}\Big)^\alpha < 0. \quad (55)$$

The costate is *negative* for high types and decays to zero at the Pareto rate $z^{-\alpha}$. Negative $\lambda(z)$ means the value function V is *decreasing* in the inherited utility $\mathcal{U}$: giving more utility to a high-productivity type is socially costly because it tightens the incentive constraint for all lower types, requiring the planner to give them more utility to maintain IC.

Remark 11 (Negative costate and redistribution)**.** *In the bounded-support problem, $\lambda(z) \geq 0$ everywhere under redistributive weights (the costate equals the redistribution pressure from above, which is non-negative). With Pareto unbounded support and zero tail weight, $\lambda(z) < 0$ for z above the crossing point $\psi(z^{**}) = 1$. The sign reversal signals that the planner is willing to earn revenue from top types at the cost of distorting their labor supply—exactly what a positive top marginal tax rate achieves.*

B.5 The Diamond–Saez formula from the asymptotic HJB

Substituting the asymptotic costate (55) into the tax formula derived from the HJB first-order condition (the general formula (41) applied to the quasilinear case with constant elasticity $e = 1/\gamma$):

$$\begin{aligned} \frac{T'}{1 - T'} &= -\frac{1 + \varepsilon^u}{\varepsilon^c} \cdot \frac{\lambda(z)}{z\, f(z)\, \mu} \\ &\approx \frac{1 + \gamma}{} \cdot \frac{1 - F(z)}{z\, f(z)} \\ &= \frac{1 + \gamma}{\alpha}, \end{aligned} \quad (56)$$

where the second line uses $\lambda(z)/\mu \approx -(1 - F(z))$ and the third uses the Pareto inverse hazard rate (51). Since $(1+\varepsilon^u)/\varepsilon^c = (1+1/e)/(1/e) = 1+\gamma$ for quasilinear preferences,

and converting to the earnings Pareto parameter via $\alpha = \alpha_y(1+e)/e = \alpha_y(1+\gamma)$:

$$\frac{T'_\infty}{1-T'_\infty} = \frac{1+\gamma}{\alpha_y(1+\gamma)} = \frac{1}{\alpha_y\, e}, \tag{57}$$

giving the Diamond–Saez formula:

$$\boxed{T'_\infty = \frac{1}{1+\alpha_y\, e}.} \tag{58}$$

B.6 Interpretation through the HJB lens

Formula (58) can be read directly from three elements of the HJB:

(i) **Costate decay rate** (α)**.** $-\lambda(z)/[1-F(z)] \to 1$: the costate decays at exactly the rate of the Pareto tail. A heavier tail (α smaller) means the costate $\lambda(z)$ is numerically large relative to density, so the incentive-constraint distortion is large and the tax rate is high.

(ii) **Inverse hazard rate** ($1/\alpha$)**.** The Pareto inverse hazard rate (51) is the ratio $(1-F)/(zf)$ that converts the magnitude of λ per person into the tax wedge per unit of income. With a thicker tail (smaller α), more types are affected by any given marginal rate, so the revenue gain from a rate increase is large relative to the behavioral distortion at the margin.

(iii) **Elasticity** (e)**.** The factor $(1+\varepsilon^u)/\varepsilon^c = (1+\gamma)$ multiplies the inverse hazard rate. A more elastic labor supply (smaller $1/\gamma$, i.e., larger e) amplifies the behavioral revenue loss from a rate increase, pushing the optimal rate down.

The HJB perspective makes these three forces transparent: they are the gradient of the value function (λ), the geometry of the type distribution (f and F), and the curvature of the flow payoff (the elasticity terms in the maximand).

B.7 Zero top rate under bounded support vs. positive top rate under Pareto: a unified view

In both cases the tax formula is $T'/(1-T') = -(1+\varepsilon^u)/\varepsilon^c \cdot \lambda(z)/(zf(z)\mu)$. The difference is entirely in the behavior of $\lambda(z)/[zf(z)]$:

- **Bounded support:** $1 - F(z) \to 0$ as $z \to \bar{z}$ faster than density $f(z) > 0$, so $(1 - F(z))/(z\, f(z)) \to 0$. Combined with $\lambda(\bar{z}) = 0$, the tax formula gives $T'(\bar{z}) = 0$.

- **Pareto tail:** $(1 - F(z))/(z\, f(z)) \to 1/\alpha > 0$ and $\lambda(z) \approx -(1 - F(z))$, so the ratio $\lambda(z)/(zf(z)) \to -1/\alpha \neq 0$. The tax formula gives $T'_\infty = 1/(1 + \alpha_y e) > 0$.

The zero-rate result and the Diamond–Saez formula are thus two limiting cases of the same HJB first-order condition, differing only in the asymptotic behavior of the Pareto inverse hazard rate.

B.8 Asymptotic value function under Pareto

Under the linear guess the value function is $V(u, z; \mu) = \lambda(z)\, u + K(z; \mu)$. For large z:

$$\lambda(z) \approx -(z_0/z)^\alpha, \qquad K(z; \mu) \approx -\int_z^\infty \mathcal{F}(\zeta)\, d\zeta \tag{59}$$

where $\mathcal{F}(\zeta) = \mu[y^*(\zeta) - C(\mathcal{U}^*(\zeta), y^*(\zeta))]f(\zeta)$ is the flow resource surplus at ζ. Both $\lambda(z)$ and $K(z; \mu)$ decay polynomially at rate $z^{-\alpha}$, consistent with the transversality condition $V(u, z; \mu) \to 0$. The rate of decay is the Pareto shape α: heavier tails (smaller α) slow the decay of the continuation value, put more weight on incentive constraints at high types, and push up the optimal tax rate.

B.9 Table: bounded support vs. Pareto tail

Remark 12 (Empirical calibration)**.** *The Diamond–Saez formula* (58) *requires two empirical inputs: α_y (the Pareto shape of the earnings distribution) and e (the compensated elasticity of taxable income). Saez (2001) surveys estimates suggesting $\alpha_y \approx 2$ for the United States, and a benchmark value of $e \approx 0.25$ for broad income, giving $T'_\infty \approx 1/(1 + 2 \times 0.25) = 2/3 \approx 67\%$. With higher elasticity, say $e = 0.5$, the formula gives $T'_\infty = 1/(1+1) = 50\%$. Diamond and Saez (2011) conclude that T'_∞ in the range* 50–70% *is consistent with the empirical evidence, well above the top marginal rates prevailing in most countries at the time.*

Table 2: Mirrlees HJB: Bounded vs. Pareto-tailed productivity

Bounded support $\mathcal{Z} = [z_0, \bar{z}]$	**Pareto tail** $\mathcal{Z} = [z_0, \infty)$
Upper BC: $\lambda(\bar{z}) = 0$ (point condition)	Upper BC: $\lambda(\infty) = 0$ (transversality)
$(1 - F(\bar{z}))/(z_{\bar{z}} f(\bar{z})) = 0$	$(1 - F(z))/(zf(z)) \to 1/\alpha > 0$
$\lambda(z)/[zf(z)] \to 0$ as $z \to \bar{z}$	$\lambda(z)/[zf(z)] \to -1/\alpha < 0$
Zero top rate: $T'(\bar{z}) = 0$	Positive top rate: $T'_\infty = 1/(1 + \alpha_y e) > 0$
Value fn. decay: $V(u, z) \to 0$ at finite $\bar{z}$	Value fn. decay: $V(u, z) \sim z^{-\alpha} \to 0$
Costate: $\lambda(z) \geq 0$ is hump-shaped	Costate: $\lambda(z)$ turns negative for large z

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